

Triangle Centers Relative to Their Cevian, Circumcevian, and Pedal Triangles

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August 2026

Abstract

For a triangle center $P = X_i(ABC)$, we ask whether the same Euclidean point is classified as X_j relative to the cevian, circumcevian, or pedal triangle of P . Circumcevian–pedal similarity and its isogonal transfer principle are classical; we give a self-contained coordinate proof and use them as the organizing mechanism. The new algebraic contribution is an oriented parameter-reversal theorem for the family F_λ , including an explicit branch discriminator. It yields a general near-equilateral obstruction to positive-Heron fixed-point claims and explains seven numerical candidates that survive broad generic testing but fail near the equilateral locus. We also prove the newly identified fixed relation $M_2 : X_{67371} \mapsto X_{67371}$ by affine covariance of two symmetric polynomials. Classical relations involving the Steiner-inellipse foci and X_{187} are recovered by short intrinsic proofs rather than claimed as new.

Using a local snapshot of the *Encyclopedia of Triangle Centers* (ETC) contiguous through X_{72807} , we enumerate every diagonal index and the two directed off-diagonal strips $\min(i, j) \leq 1000$. Of 161 discoveries, 54 pass a 60-digit audit on 36 labelled instances of six scalene shapes. A 70-digit test on 40 additional shapes rejects 13, and the exact near-equilateral analysis rejects seven more, leaving 34 rows not eliminated by these tests. Exact arguments in this paper prove 15 of these rows, while three further rows are classical; the remaining 16 are unresolved computational candidates. Among the rejected relations are $M_2 : X_{485} \mapsto X_{61083}$ and $M_3 : X_{485} \mapsto X_{61084}$. Finally, X_{795} is proved to lie on the circumcircle, so its circumcevian triple is degenerate. The remaining rows not covered by the exact theorems are explicitly retained only as computational candidates.

Keywords. triangle center; cevian triangle; circumcevian triangle; pedal triangle; experimental geometry; Encyclopedia of Triangle Centers.

1 The geometric question

Let ABC be a labelled, nondegenerate scalene triangle and let P be a finite point distinct from the vertices. Denote by

- $\mathcal{T}_1(P) = A_1B_1C_1$ the cevian triangle, with $A_1 = AP \cap BC$ and cyclically;
- $\mathcal{T}_2(P) = A_2B_2C_2$ the circumcevian triangle, where A_2 is the second intersection of AP with the circumcircle and cyclically;
- $\mathcal{T}_3(P) = A_3B_3C_3$ the pedal triangle, whose vertices are the perpendicular projections of P on BC, CA, AB .

Only noncollinear derived triples are admitted. We write

$$\mathcal{C}_k(i, j; ABC) : \quad X_i(ABC) = X_j(\mathcal{T}_k(X_i(ABC))). \quad (1)$$

Thus $M_k : X_i \mapsto X_j$ records a reclassification of the *same Euclidean point in the ambient plane*; it is not an ordinary point map. The case $i = j$ is called a fixed-point candidate.

Figure 1 summarizes the three changes of reference triangle. It is schematic rather than metrically exact.

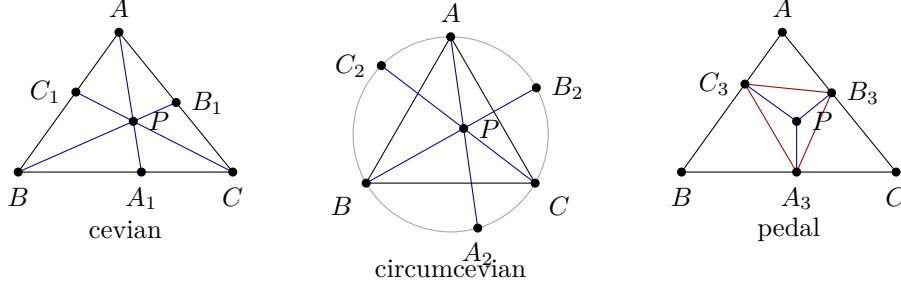


Figure 1: The three derived reference triangles in (1).

If P is on the circumcircle, $\mathcal{T}_2(P)$ collapses to P, P, P , while $\mathcal{T}_3(P)$ is collinear by the Simson theorem. Neither configuration admits an ETC target center. Selected-root definitions may also change branch as the triangle shape or its labelling changes. These facts motivate the degeneracy checks and multiple-shape audits below.

2 Classical transfer principle and exact construction

Let $P = (u : v : w)$ in barycentric coordinates of ABC , with side lengths $a = BC$, $b = CA$, $c = AB$. The cevian traces are

$$A_1 = (0 : v : w), \quad B_1 = (u : 0 : w), \quad C_1 = (u : v : 0).$$

The circumcircle has barycentric equation

$$a^2yz + b^2zx + c^2xy = 0.$$

Intersecting this equation with AP and removing the known root A gives

$$A_2 = (a^2vw : -v(b^2w + c^2v) : -w(b^2w + c^2v)), \quad (2)$$

with B_2, C_2 obtained cyclically. The implementation constructs all three derived triangles independently in Cartesian coordinates; equation (2) is an exact audit of the line-circle calculation.

The circumcevian and pedal problems are not independent. The following theorem provides the algebraic bridge between them.

Theorem 1 (Circumcevian–pedal isogonal transfer). *Let $P = (u : v : w)$ be admissible, with $uvw(u + v + w) \neq 0$, and suppose that $\mathcal{T}_2(P)$ and $\mathcal{T}_3(P)$ are nondegenerate. Under the correspondence $A_2 \leftrightarrow A_3$, $B_2 \leftrightarrow B_3$, $C_2 \leftrightarrow C_3$, the two derived triangles have proportional corresponding squared side lengths. Moreover, the barycentric coordinate triples of the same point P in these triangles are isogonal conjugates. Consequently, for every side-defined center function X_j on a consistent branch,*

$$P = X_j(\mathcal{T}_2(P)) \iff P = \iota(X_j)(\mathcal{T}_3(P)). \quad (3)$$

Proof. Put $A = a^2$, $B = b^2$, $C = c^2$ and $t = u + v + w$. Let R_A, R_B, R_C denote the three raw triples supplied by (2), and let s_A, s_B, s_C be their coordinate sums. Direct multiplication gives

$$uR_A + vR_B + wR_C = -(Avw + Bwu + Cuv)(u, v, w).$$

Since $A_2 = R_A/s_A$ and cyclically, the coordinates of P relative to its circumcevian triangle are therefore

$$P_{\mathcal{T}_2} = (us_A : vs_B : ws_C). \quad (4)$$

The barycentric distance formula also gives

$$PA^2 = -\frac{s_A}{t^2}, \quad PB^2 = -\frac{s_B}{t^2}, \quad PC^2 = -\frac{s_C}{t^2}.$$

Thus (4) may be written

$$P_{\mathcal{T}_2} = (uPA^2 : vPB^2 : wPC^2). \quad (5)$$

For comparison, the foot on BC is

$$A_3 = \frac{1}{2At}(0, u(A+B-C) + 2Av, u(A+C-B) + 2Aw),$$

with the other feet obtained cyclically. Inverting this affine matrix gives the particularly simple triple

$$P_{\mathcal{T}_3} = \left(\frac{A}{u} : \frac{B}{v} : \frac{C}{w} \right). \quad (6)$$

Let $a_2 = B_2C_2$, $a_3 = B_3C_3$, let $\mathcal{R}$ be the circumradius of ABC , and let Π_P be the signed power of P with respect to its circumcircle. The projection formula and the power-of-a-point theorem give

$$a_3^2 = \frac{a^2 PA^2}{4\mathcal{R}^2}, \quad a_2^2 = \frac{\Pi_P^2 a^2}{PB^2 PC^2}.$$

Hence

$$\frac{a_2^2}{a_3^2} = \frac{4\mathcal{R}^2 \Pi_P^2}{PA^2 PB^2 PC^2},$$

and the same factor occurs cyclically. The corresponding side triples are therefore proportional. Isogonal conjugation of (5) inside $A_2B_2C_2$ yields

$$\left(\frac{a_2^2}{uPA^2} : \frac{b_2^2}{vPB^2} : \frac{c_2^2}{wPC^2} \right) \sim \left(\frac{A}{u} : \frac{B}{v} : \frac{C}{w} \right),$$

which is exactly (6). This proves the assertion. $\square$

3 Relation to previous circumcevian theory

The similarity of the circumcevian and pedal triangles under the natural vertex correspondence is classical; it appears in Kimberling's treatment of central triangles [2]. Stothers gave the corresponding coordinate transfer theorem and tables of circumcevian classes [3, 4]. In particular, those tables already contain the reciprocal centroid–Steiner-focus class, the invariant centers X_3, X_6, X_{187} , the class $X_{36} \leftrightarrow X_{186}$, and the one-parameter extraversion class

$$\{Q(\phi), Q(-\phi)\}, \quad Q(\phi) = (\sin A \sin(A + \phi) : \sin B \sin(B + \phi) : \sin C \sin(C + \phi)).$$

Up to a projective scalar this is the present F_λ family with $\lambda = -\cot \phi$. Gibert's related tables also organize centers with trilinears of the form $\tan(A + t)$ and associated Brocard-axis families [5].

Accordingly, Theorem 1, the Steiner-focus result, and the X_{187} relations below are included as self-contained recoveries and as checks on notation. The contribution developed here is more specific: it distinguishes the signed, oriented identity from the usual positive-Heron convention, gives an exact formula for the branch change, uses it to disprove apparent global fixed points, proves the high-index fixed point X_{67371} , and audits a modern ETC snapshot far beyond the classical tables.

4 The exact F_λ family and its branch locus

4.1 The oriented parameter-reversal law

Put

$$H = \sqrt{(a+b-c)(a-b+c)(-a+b+c)(a+b+c)} = 4[ABC]$$

and

$$D_a = a^2 - b^2 - c^2, \quad D_b = b^2 - c^2 - a^2, \quad D_c = c^2 - a^2 - b^2.$$

Define

$$F_\lambda = (a^2(D_a + \lambda H) : b^2(D_b + \lambda H) : c^2(D_c + \lambda H)). \quad (7)$$

Its trilinears are

$$(-\cos A + \lambda \sin A : -\cos B + \lambda \sin B : -\cos C + \lambda \sin C).$$

For an oriented labelled triangle $T = UVW$, write $H_T^{\text{or}} = 2 \det(V - U, W - U)$, and use a superscript “or” when this signed radical replaces the positive Heron radical.

Theorem 2 (Circumcevian parameter reversal). *On the generic nondegenerate domain,*

$$F_\lambda(ABC) = F_{-\lambda}^{\text{or}}(\mathcal{T}_2(F_\lambda(ABC))). \quad (8)$$

The identity extends after cancellation wherever both sides remain defined. The exceptional loci include $3\lambda^2 = 1$ and the vanishing factors displayed in the proof.

Proof. Set $A = a^2$, $B = b^2$, $C = c^2$, $S = A + B + C$; then

$$H^2 = 2AB + 2BC + 2CA - A^2 - B^2 - C^2, \quad D_a = 2A - S.$$

Define cyclically

$$q_a = (3\lambda^2 - 1)A - 2\lambda^2 S + 2\lambda H, \quad Q = (3\lambda^2 + 1)H - 2\lambda S. \quad (9)$$

Substitution of $u = A(D_a + \lambda H)$ and its cyclic companions into (2) shows that the coordinate sums of the raw circumcevian triples are

$$s_A = BCH^2 q_a, \quad s_B = CAH^2 q_b, \quad s_C = ABH^2 q_c.$$

By (4), the original point relative to its circumcevian triangle therefore has coordinates

$$((D_a + \lambda H)q_a : (D_b + \lambda H)q_b : (D_c + \lambda H)q_c). \quad (10)$$

Let $A' = a_2^2$, $B' = b_2^2$, $C' = c_2^2$. Exact reduction of the barycentric distance formula gives

$$A' = \delta q_a, \quad B' = \delta q_b, \quad C' = \delta q_c, \quad \delta = \frac{ABCQ^2}{H^2 q_a q_b q_c}. \quad (11)$$

The determinant of the normalized circumcevian triples gives

$$H_2^{\text{or}} = -\delta Q. \quad (12)$$

Equivalently, its square follows from the factorization

$$2(q_a q_b + q_b q_c + q_c q_a) - q_a^2 - q_b^2 - q_c^2 = Q^2.$$

Finally,

$$q_a - q_b - q_c + \lambda Q = (3\lambda^2 - 1)(D_a + \lambda H), \quad (13)$$

cyclically. The first coordinate of $F_{-\lambda}^{\text{or}}$ in the derived triangle is, by (11)–(12),

$$A'(A' - B' - C' - \lambda H_2^{\text{or}}) = \delta^2 q_a (q_a - q_b - q_c + \lambda Q).$$

Equation (13) makes this proportional to the first coordinate in (10); the other two coordinates follow cyclically. $\square$

The theorem contains, on the corresponding oriented ETC branches,

$$X_3 = F_0, \quad X_{61} = F_{-\sqrt{3}}, \quad X_{62} = F_{\sqrt{3}}, \quad X_{371} = F_{-1}, \quad X_{372} = F_1,$$

and the limiting member $F_\infty = X_6$. It therefore explains the circumcevian rows $X_3 \mapsto X_3$, $X_6 \mapsto X_6$, $X_{61} \mapsto X_{62}$, and $X_{371} \mapsto X_{372}$.

Let ι denote isogonal conjugation and put

$$G_\lambda = \iota(F_\lambda) = \left(\frac{1}{D_a + \lambda H} : \frac{1}{D_b + \lambda H} : \frac{1}{D_c + \lambda H} \right). \quad (14)$$

Corollary 3 (Pedal parameter law). *On the same generic oriented domain,*

$$F_{-\lambda}(ABC) = G_\lambda^{\text{or}}(\mathcal{T}_3(F_{-\lambda}(ABC))). \quad (15)$$

Proof. Apply Theorem 2 to $P = F_{-\lambda}$ and then apply Theorem 1. $\square$

This proves the corresponding pedal rows $X_3 \mapsto X_4$, $X_6 \mapsto X_2$, $X_{61} \mapsto X_{18}$, and $X_{371} \mapsto X_{486}$ on their stated branches.

Corollary 4 (Recovery of the classical X_{187} relations). *Where the constructions are defined,*

$$M_2 : X_{187} \mapsto X_{187}, \quad M_3 : X_{187} \mapsto X_{671}.$$

Proof. The ETC formula reduces exactly to $X_{187} = F_\lambda$ with $\lambda = S/(3H)$. From (9),

$$S_q = q_a + q_b + q_c = 6\lambda H - (3\lambda^2 + 1)S,$$

and direct subtraction gives

$$HS_q - SQ = 2(H - \lambda S)(3H\lambda - S).$$

Thus $S_q/Q = S/H$ at $\lambda = S/(3H)$. Equations (11) and (12) show that the oriented parameter of the derived triangle is $-\lambda$; after returning to the positive-Heron convention, the center is again X_{187} . The pedal result follows from Theorem 1 and the ETC identity $\iota(X_{187}) = X_{671}$. $\square$

4.2 Branch locus and constant-family obstruction

The positive-Heron convention can hide the parameter reversal. From (12),

$$H_2^{\text{or}} = -\frac{ABCQ^3}{H^2q_aq_bq_c}. \quad (16)$$

If this quantity is negative, $F_{-\lambda}^{\text{or}}$ is represented as F_λ in the positive-Heron convention, producing an apparent fixed point. If it is positive, the positive-Heron target is instead $F_{-\lambda}$.

The phenomenon is systematic, rather than peculiar to the seven ETC entries found by the search. At the equilateral point $A = B = C = x$, (9) reduces to

$$q_a = q_b = q_c = -(\sqrt{3}\lambda - 1)^2x, \quad Q = \sqrt{3}x(\sqrt{3}\lambda - 1)^2. \quad (17)$$

Consequently, for every finite real $\lambda \neq 1/\sqrt{3}$, equation (16) is positive at the equilateral limit and remains positive throughout a sufficiently small scalene neighbourhood.

Moreover, if $\lambda \neq 0$ and a generic scalene triangle satisfied $F_\lambda = F_{-\lambda}$, comparison of the first two projective coordinates would give

$$(D_a + \lambda H)(D_b - \lambda H) = (D_b + \lambda H)(D_a - \lambda H),$$

and hence $D_a = D_b$. Cyclic comparison gives $D_a = D_b = D_c$, which forces $a = b = c$. Thus in the scalene neighbourhood the positive-Heron target $F_{-\lambda}$ is genuinely different from the source F_λ .

The omitted value $\lambda = 1/\sqrt{3}$ is not a hidden global fixed point. Here $3\lambda^2 - 1 = 0$, so $q_a = q_b = q_c$ for every triangle on the regular part of the construction. Equation (11) makes the circumcevian triangle equilateral, while the positive-Heron formula for $F_{1/\sqrt{3}}$ on that triangle is $(0 : 0 : 0)$ and therefore defines no target center. We have proved the following global obstruction.

Corollary 5 (Constant-family obstruction). *No finite nonzero constant member F_λ defines an unrestricted fixed point under the standard positive-Heron convention. Within the oriented parameter-reversal law, $F_0 = X_3$ is fixed, and $F_\infty = X_6$ is the corresponding limiting fixed member.*

Exact substitution in (16) is negative for $(4, 6, 7)$ but positive for the scalene near-equilateral triangle $(100000, 100001, 100002)$ for every row in Table 1. The ETC formulas in the table reduce exactly to the displayed constant F_λ . Hence all seven former fixed-point candidates fail on the near-equilateral component as immediate instances of the corollary.

Table 1: Additional M_2 fixed-point candidates rejected exactly by the near-equilateral branch test.

X_{6408}	X_{6454}	X_{6469}	X_{6481}	X_{10140}	X_{10144}	X_{10146}
$4/7$	$3/5$	$7/12$	$5/9$	$25/44$	$120/209$	$16/29$

For example, $F_{4/7} = X_{6408}$ is classified there as $F_{-4/7} = X_{6407}$, not as X_{6408} . These parameters cluster near $1/\sqrt{3}$, where the branch boundary approaches the equilateral point; this explains why broad random tests can miss the thin component.

4.3 The fixed point X_{67371}

Theorem 6 (The X_{67371} fixed point). *Subject to the usual nondegeneracy and definition qualifications,*

$$M_2 : X_{67371} \mapsto X_{67371}$$

is an exact identity.

Proof. Write

$$\sigma_1 = A + B + C, \quad \sigma_2 = AB + BC + CA, \quad \sigma_3 = ABC.$$

Define

$$U = \frac{3}{2}(3A - \sigma_1)(3B - \sigma_1)(3C - \sigma_1) = \frac{3}{2}(2\sigma_1^3 - 9\sigma_1\sigma_2 + 27\sigma_3)$$

and

$$W = -\sigma_1^4 + \frac{15}{2}\sigma_1^2\sigma_2 + \frac{27}{2}\sigma_1\sigma_3 - 18\sigma_2^2.$$

If P_a is the degree-four factor multiplying $a^2 = A$ in the ETC barycentrics of X_{67371} , exact expansion gives

$$P_a = U(2A - \sigma_1) + W,$$

cyclically. Hence

$$X_{67371} = F_\Lambda, \quad \Lambda = \frac{W}{UH}. \tag{18}$$

The mechanism is affine covariance, not a fortuitous large factorization. Put

$$k = 3\lambda^2 - 1, \quad n = 2\lambda(H - \lambda\sigma_1).$$

Equation (9) is simply $q_i = kA_i + n$. Let U_q, W_q denote the same symmetric polynomials evaluated on (q_a, q_b, q_c) . Because

$$3q_a - (q_a + q_b + q_c) = k(3A - \sigma_1),$$

and cyclically, one immediately obtains

$$U_q = k^3 U. \tag{19}$$

Substitution in the symmetric expression for W (or direct collection by degree in the translation n) gives the companion covariance

$$W_q = k^3(kW + nU). \tag{20}$$

Now set $\lambda = \Lambda = W/(UH)$. Then

$$\frac{W_q}{U_q} = k \frac{W}{U} + n = k\lambda H + n = \lambda((3\lambda^2 + 1)H - 2\lambda\sigma_1) = \lambda Q.$$

Homogeneity, together with (11) and (12), therefore gives

$$\Lambda(\mathcal{T}_2(P)) = -\frac{W_q}{U_q Q} = -\lambda.$$

Thus X_{67371} evaluated in the circumcevian triangle is precisely $F_{-\lambda}^{\text{or}}$, which equals the original point by Theorem 2. $\square$

5 Centroid preimages and an exact degeneracy theorem

Theorem 7 (Steiner-inellipse focus characterization; classical). *Let P lie off the circumcircle and suppose its circumcevian triangle is nondegenerate. Then*

$$P \text{ is the centroid of } \mathcal{T}_2(P) \iff P \text{ is a focus of the Steiner inellipse of } ABC.$$

Consequently,

$$M_2 : X_{39162} \mapsto X_2, \quad M_2 : X_{39163} \mapsto X_2,$$

and, by Theorem 1,

$$M_3 : X_{39162} \mapsto X_6, \quad M_3 : X_{39163} \mapsto X_6.$$

This equivalence, including the ETC labels of the two foci, belongs to the classical circumcevian class table [4]. The proof below is included because it is short, intrinsic, and independently validates the coordinate pipeline used in the census.

Proof. Normalize the circumcircle to the complex unit circle, with vertex affixes a, b, c , and let p be the affix of P , $q = \bar{p}$. The second intersection of AP with the circle is

$$a_2 = \frac{p - a}{1 - qa} = p + \frac{a(pq - 1)}{1 - qa},$$

cyclically. Since $pq \neq 1$, the centroid condition $a_2 + b_2 + c_2 = 3p$ is equivalent to

$$\frac{a}{1 - qa} + \frac{b}{1 - qb} + \frac{c}{1 - qc} = 0.$$

Writing

$$\Phi(q) = (1 - aq)(1 - bq)(1 - cq) = 1 - s_1q + s_2q^2 - s_3q^3,$$

this equation is $\Phi'(q) = 0$. Conjugation, using $|a| = |b| = |c| = 1$, gives

$$3p^2 - 2s_1p + s_2 = 0.$$

This is precisely the derivative of $(z - a)(z - b)(z - c)$. The Siebeck–Marden theorem identifies its two roots with the foci of the Steiner inellipse [7, 8]. The ETC labels of these foci are X_{39162} and X_{39163} ; the final two relations follow because $\iota(X_2) = X_6$. $\square$

Together with the classical cevian and pedal cases, the nondegenerate centroid-preimage list found by the scan is therefore

$$C_1(2, 2), \quad C_2(39162, 2), \quad C_2(39163, 2), \quad C_3(6, 2),$$

and the two nonclassical circumcevian rows are now proved intrinsically.

Proposition 8 (The exact X_{795} degeneracy). *Where its barycentric formula is finite, X_{795} lies on the circumcircle of ABC . If it differs from A, B, C , then $\mathcal{T}_2(X_{795}) = (X_{795}, X_{795}, X_{795})$. Hence the formal relation $M_2 : X_{795} \mapsto X_2$ is degenerate and inadmissible.*

Proof. Put $A_0 = a^3$, $B_0 = b^3$, $C_0 = c^3$. The ETC formula is cyclically

$$X_{795} = (f_a : f_b : f_c), \quad f_a = \frac{a^2}{(B_0 - C_0)(A_0^2 - B_0C_0)}.$$

This is the published factorization because $(b - c)(b^2 + bc + c^2) = b^3 - c^3$ and $(a^2 - bc)(a^4 + a^2bc + b^2c^2) = a^6 - b^3c^3$. Substitution into the circumcircle equation and clearing the common denominator leaves

$$(B_0 - C_0)(A_0^2 - B_0C_0) + (C_0 - A_0)(B_0^2 - C_0A_0) + (A_0 - B_0)(C_0^2 - A_0B_0) = 0.$$

Thus $a^2f_bf_c + b^2f_cf_a + c^2f_af_b = 0$ identically. Every second intersection is therefore X_{795} , and the derived triple has zero area. $\square$

6 Computational protocol

The high-speed discovery comparison uses normalized barycentric triples q, r and the projective residual

$$\rho(q, r) = \frac{\|q \times r\|}{\|q\|\|r\|}, \quad (21)$$

with threshold 10^{-9} . Discovery candidates are recomputed from the symbolic ETC definitions. For a Cartesian source point Q and its proposed target $X_j(UVW)$, the audit residual is

$$\varepsilon(Q; U, V, W, j) = \frac{\|Q - X_j(UVW)\|}{\text{per}(ABC)}. \quad (22)$$

The primary audit uses 60-digit working precision and accepts only $\varepsilon < 10^{-30}$. The independent generic-shape stress test uses 70-digit precision and the stricter threshold 10^{-40} . These are the only validation protocols used for the results stated in this article.

The primary audit uses all six labellings of the acute shapes

$$(4, 6, 7), \quad (13, 20, 21), \quad (29, 30, 31)$$

and of the obtuse shapes

$$(5, 6, 8), \quad (4, 11, 13), \quad (5, 7, 10).$$

Thus a candidate receives 36 labelled tests but only six geometrically independent shapes. A row passing this stage is therefore called a *six-shape audit survivor*, not a universal identity.

7 Dataset, enumerated region, and evaluable coverage

The formula and barycentric-fingerprint snapshots each contain exactly 72807 entries, indexed consecutively from 1 to 72807. The cache builder found no index gap and no symbolic expansion failure. This database completeness is not the same as numerical evaluability: on a particular triangle a formula may be complex, infinite, indeterminate, branch-incompatible, or may produce a degenerate derived triple. An unavailable comparison is excluded from evaluable coverage and is never recorded as a disproved equality.

The nominal index region is

$$\begin{aligned} i = j, \quad 1 \leq i \leq 72807, \\ 1 \leq i, j \leq 72807, \quad i \neq j, \quad \min(i, j) \leq 1000. \end{aligned} \quad (23)$$

The off-diagonal part is the union of two directed strips, not a rectangle. It contains

$$3\{1000(72807 - 1) + (72807 - 1000)1000\} = 433839000$$

nominal comparisons per discovery seed; the diagonal contains $3 \cdot 72807 = 218421$. Discovery was performed independently on the obtuse triangle (6, 9, 13) and the acute triangle (4, 6, 7).

Table 2: Evaluable coverage of the nominal region (23).

seed	strip valid	strip coverage	diagonal valid	diagonal coverage
obtuse (6, 9, 13)	388166091	89.4724%	210351	96.3053%
acute (4, 6, 7)	388285438	89.4999%	210448	96.3497%

The two discovery seeds produced 161 distinct rows: 113 diagonal and 48 off-diagonal. The exclusions represented by Table 2 are a substantive limitation. The calculation exhaustively enumerates the stated indices, but it does not assign a true or false value to every nominal pair.

8 Results of the six-shape audit

Exactly 54 of the 161 discoveries pass all 36 labelled instances: 29 are diagonal and 25 off-diagonal. The diagonal list is

$$\begin{aligned} M_1 : & X_2, X_{13}; \\ M_2 : & X_3, X_6, X_{13}, X_{187}, X_{368}, X_{3513}, X_{3514}, X_{5467}, X_{6104}, \\ & X_{6408}, X_{6440}, X_{6454}, X_{6469}, X_{6477}, X_{6481}, X_{6522}, X_{8456}, X_{9181}, \\ & X_{10138}, X_{10140}, X_{10142}, X_{10144}, X_{10146}, X_{17851}, X_{17852}, X_{21906}, X_{67371}; \\ M_3 : & \text{none.} \end{aligned} \quad (24)$$

Here a displayed center under M_k means the diagonal candidate $M_k : X_i \mapsto X_i$. The off-diagonal list is

$$\begin{aligned} M_1 : & X_7 \mapsto X_6; \\ M_2 : & X_1 \mapsto X_4, X_{36} \mapsto X_{186}, X_{61} \mapsto X_{62}, X_{202} \mapsto X_{34321}, \\ & X_{203} \mapsto X_{34322}, X_{371} \mapsto X_{372}, X_{485} \mapsto X_{61083}, \\ & X_{34321} \mapsto X_{202}, X_{39162} \mapsto X_2, X_{39163} \mapsto X_2; \\ M_3 : & X_1 \mapsto X_3, X_3 \mapsto X_4, X_6 \mapsto X_2, X_{13} \mapsto X_{15}, \\ & X_{36} \mapsto X_{265}, X_{61} \mapsto X_{18}, X_{187} \mapsto X_{671}, \\ & X_{202} \mapsto X_{6671}, X_{203} \mapsto X_{6672}, X_{371} \mapsto X_{486}, \\ & X_{485} \mapsto X_{61084}, X_{30335} \mapsto X_{175}, X_{39162} \mapsto X_6, X_{39163} \mapsto X_6. \end{aligned} \quad (25)$$

Equations (24) and (25) report a finite numerical audit. They are not theorem statements.

The M_2 and M_3 rows in (25) must not be counted as independent discoveries. The classical transfer principle pairs an M_2 target X_j with the isogonal target $\iota(X_j)$ under M_3 . Thus, for example,

$$\begin{array}{lll}
M_2 : X_{36} \mapsto X_{186} & \iff & M_3 : X_{36} \mapsto X_{265}, \\
M_2 : X_{61} \mapsto X_{62} & \iff & M_3 : X_{61} \mapsto X_{18}, \\
M_2 : X_{202} \mapsto X_{34321} & \iff & M_3 : X_{202} \mapsto X_{6671}, \\
M_2 : X_{203} \mapsto X_{34322} & \iff & M_3 : X_{203} \mapsto X_{6672}, \\
M_2 : X_{371} \mapsto X_{372} & \iff & M_3 : X_{371} \mapsto X_{486}, \\
M_2 : X_{485} \mapsto X_{61083} & \iff & M_3 : X_{485} \mapsto X_{61084}, \\
M_2 : X_{39162}, X_{39163} \mapsto X_2 & \iff & M_3 : X_{39162}, X_{39163} \mapsto X_6.
\end{array}$$

Both manifestations are retained in the numerical census for reproducibility, but each line represents one structural coincidence.

The new high-index diagonal relation $M_2 : X_{67371} \mapsto X_{67371}$ and the two forward candidates from X_{485} first appear when the opposite or diagonal bound is extended beyond 53700. Four reverse rows show a sharp split in the primary audit:

$$\begin{array}{ll}
M_2 : X_{61083} \mapsto X_{485}, & X_{61084} \mapsto X_{486}, \\
M_3 : X_{61083} \mapsto X_{371}, & X_{61084} \mapsto X_{372}.
\end{array} \tag{26}$$

Each passes all 18 acute labelled instances and none of the 18 obtuse instances. This is branch evidence, not an acute-triangle theorem.

9 Independent generic-shape stress test

To probe the two-dimensional triangle-shape domain more broadly, the 54 six-shape survivors were tested, without further discovery, on 40 additional integer-sided scalene shapes. A fixed Mersenne-Twister seed, 20260831, generated 20 acute and 20 obtuse shapes; the exact ordered triples are stored with the output. All $54 \cdot 40 = 2160$ comparisons were evaluable. There were 2111 passes and 49 failures. Thus 41 candidates pass all 40 additional shapes and 13 do not.

Table 3: Six-shape survivors rejected by the 40-shape stress test. A and O denote acute and obtuse additional shapes.

map	candidate	passes	failures	location of failures
M_1	$X_{13} \mapsto X_{13}$	32	8	8 O
M_2	$X_{13} \mapsto X_{13}$	32	8	8 O
M_2	$X_{485} \mapsto X_{61083}$	38	2	2 O
M_2	$X_{6104} \mapsto X_{6104}$	32	8	8 O
M_2	$X_{6440} \mapsto X_{6440}$	39	1	1 A
M_2	$X_{6477} \mapsto X_{6477}$	39	1	1 A
M_2	$X_{6522} \mapsto X_{6522}$	39	1	1 A
M_2	$X_{8456} \mapsto X_{8456}$	33	7	4 A, 3 O
M_2	$X_{10138} \mapsto X_{10138}$	39	1	1 A
M_2	$X_{10142} \mapsto X_{10142}$	39	1	1 A
M_2	$X_{34321} \mapsto X_{202}$	39	1	1 O
M_3	$X_{13} \mapsto X_{15}$	32	8	8 O
M_3	$X_{485} \mapsto X_{61084}$	38	2	2 O

Both X_{485} candidates fail on the ordered obtuse shapes (60, 33, 90) and (83, 34, 50). Including the primary audit, each therefore passes 74 of 76 labelled instances, but neither may be

advertised as an unrestricted identity. By contrast, $M_2 : X_{67371} \mapsto X_{67371}$ passes all 76 stated instances and is proved exactly by Theorem 6.

The 41 random-test survivors are not the final unrestricted list. The exact family analysis in Section 4 supplies near-equilateral counterexamples to seven more fixed-point candidates. Thus 20 of the 54 six-shape survivors are now rejected and 34 rows remain after all stated tests. The exact arguments in this paper prove 15 of these rows, while the classical cevian-centroid identity and Stothers’s X_{36} class account for three more. Specifically, the 15 consist of the eight F_λ family rows for $X_3, X_6, X_{61}, X_{371}$, the two X_{187} rows, the X_{67371} fixed row, and the four Steiner-focus rows. The remaining 16 rows are unresolved computational candidates.

Table 3 also shows that “acute versus obtuse” is not a sufficient branch classification: several candidates fail on only one of 20 additional acute shapes, while passing the other 19. Any definitive statement must identify discriminant curves and connected components in triangle-shape space.

10 Limits of the search

For $N = 72807$, testing all three maps on every ordered source-target pair would require

$$3N^2 = 15,902,577,747$$

comparisons on one discovery triangle. The present computation exhaustively enumerates the diagonal and the two strips in (23); it does not classify off-diagonal pairs with both indices above 1000. Moreover, the coverage in Table 2 prevents a claim of complete classification even inside the nominal strips. A full search should first develop an indexed invariant, a nearest-neighbour structure for derived fingerprints, or a family-level symbolic reduction. Before merely enlarging the quadratic source–target search, the next stage should also compare every surviving class systematically with Kimberling’s central-triangle theory, Stothers’s circumcevian and pedal tables, and Gibert’s parameter families. That comparison is mathematically more informative than increasing the raw index bound without a structural invariant.

11 Reproducibility and logical status

The calculation used Wolfram 15.0.0 for Microsoft Windows (64-bit), dated May 19, 2026, and an ETC snapshot written on August 22, 2026. The complete reproducibility bundle contains the database snapshots, unified search and validation scripts, all ordered test triangles and residuals, and independent exact checks for the symbolic theorems. The machine-readable exact-check report has Boolean field `allChecksTrue` equal to `true`. Appendix A records local filenames and SHA-256 checksums; these implementation details are kept out of the geometric narrative.

Before submission, these files and the database snapshot should be deposited in a permanent public archive with a stable identifier. Local filenames and checksums improve traceability but do not replace archival access.

Theorems 1, 2, 6, and 7, their corollaries, and Proposition 8 are exact. A row not explicitly covered by these or by a cited classical argument remains a computational candidate, even if it belongs to the final set of 34 rows not eliminated by the stated tests.

Acknowledgements

This problem was initially posed by Mowaffaq Hajja in unpublished work and was communicated personally to the author. Professor Hajja generously provided collected material and conjectures concerning the problem, encouraged the investigation, and granted permission for that material to be used. The author gratefully acknowledges his generosity and sustained support.

Declaration on the use of generative AI

During the preparation of this article, the author used ChatGPT, a generative-AI system developed by OpenAI, to assist with drafting symbolic computation code, checking algebraic manipulations computationally, editing language, and preparing L^AT_EX. The author independently reviewed and verified the mathematical arguments, computations, references, and final text, and accepts full responsibility for the contents of the article. The AI system did not determine the research conclusions, does not meet the criteria for authorship, and is not a coauthor.

A Relation to the preliminary computations

The search was originally run in a first-500 block through X_{53700} and a supplementary 501–1000 block. Those checkpointed computations remain in `output/m1m2m3/offdiagonal_first500` and `output/m1m2m3/offdiagonal_501_1000`. The final counts and lists in the main text are obtained from the unified X_{72807} aggregation, with overlaps subtracted. This appendix records provenance without asking the reader to follow superseded search scopes as part of the main argument.

A.1 Local reproducibility record

The ETC formula and normalized-barycentric snapshots have respective SHA-256 checksums

```
B10F8A7FBF545F25DB2FCCC99D94022EA8325088BE15C9B06858610CE473F5C0
2DFD39CB60A600B2E06750462F6187C9B8C03F647C5F08B8EEC7E69A9C8F2D67,
```

and the expanded 72807-entry cache has checksum

```
ADD5F1F1686FB8EBC67942BA0052087CED2BA0EA0CEE4955703F705ACC622027.
```

The unified calculation is driven by

```
m1m2m3_unified_build_cache.wls, m1m2m3_unified_worker.wls,
run_m1m2m3_unified_workers.ps1,
m1m2m3_unified_validate.wls, m1m2m3_unified_aggregate.wls.
```

Canonical outputs are in `output/m1m2m3/unified`. The historical filename `universal_candidates.csv` means only “passed the six-shape audit.”

The exact X_{795} check is `m1m2m3_referee_exact_checks.wls`. The 40-shape audit and its residuals are stored under `output/m1m2m3/referee_revision`. The exact reductions for the family and X_{67371} theorems are checked by `m1m2m3_algebraic_revision_checks.wls`; its report is `output/m1m2m3/algebraic_revision/algebraic_revision_checks.json`. The root of the local LeJean–Kimberling data tree is supplied to that script as its first command-line argument or through `LEJEAN_KIMBERLING_ROOT`.

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